The documentation and process conversion measures necessary to comply with this revision shall be completed by 7 December 2001.

INCH-POUND

MIL-PRF-19500/543F
7 September 2001
SUPERSEDING
MIL-PRF-19500/543E
5 August 1997

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, FIELD EFFECT TRANSISTORS, N-CHANNEL, SILICON
REPETITIVE AVALANCHE TYPES 2N6764, 2N6766, 2N6768, 2N6770,
JAN, JANTX, JANTXV, JANS, JANHC and JANKC

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for N-channel, enhancement-mode, MOSFET, power transistors. Four levels of product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500 and two levels of product assurance for each unencapsulated die, with avalanche energy ratings ($E_{AS}$ and $E_{AR}$) and maximum avalanche current ($I_{AR}$).

1.2 Physical dimensions. See figure 1 (TO-204AE for types 2N6764 and 2N6766; TO-204AA for types 2N6768 and 2N6770 (formerly TO-3)), see figures 2 and 3 for JANHC and JANKC (die) dimensions.

1.3 Maximum ratings. ($T_A$ = +25°C, unless otherwise specified).

| Type | $P_T$ (1) $T_C$ = +25° C | $P_T$ $T_C$ = +25° C | $V_{DS}$ | $V_{DG}$ | $V_{GS}$ | $I_{D1}$ (2) $T_C$ = +25° C | $I_S$ | $I_{D2}$ (2) $T_C$ = +100° C |
|---|---|---|---|---|---|---|---|---|
| | W | W | V dc | V dc | V dc | A dc | A dc | A dc |
| 2N6764 | 150 | 4 | 100 | 100 | ± 20 | 38.0 | 38.0 | 24.0 |
| 2N6766 | 150 | 4 | 200 | 200 | ± 20 | 30.0 | 30.0 | 19.0 |
| 2N6768 | 150 | 4 | 400 | 400 | ± 20 | 14.0 | 14.0 | 9.0 |
| 2N6770 | 150 | 4 | 500 | 500 | ± 20 | 12.0 | 12.0 | 7.75 |

| Type | $I_{DM}$ (3) | $E_{AS}$ | $E_{AR}$ | $I_{AR}$ | $V_{ISO}$ 70,000 ft. attitude | $T_{STG}$ and $T_{OP}$ | Max $r_{DS(on)}$ (1); $V_{GS}$ = 10 V dc $I_D = I_{D2}$ | | $R_{\theta JC}$ max |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | $T_J$ = +25° C | $T_J$ = +150° C | |
| | A pk | A | mJ | mJ | | °C | Ω | Ω | °C/W |
| 2N6764 | 152 | 150 | 15 | 38.0 | | -55 | 0.055 | 0.105 | 0.83 |
| 2N6766 | 120 | 500 | 15 | 30.0 | | to | 0.085 | 0.170 | 0.83 |
| 2N6768 | 56 | 700 | 15 | 14.0 | 400 | +150 | 0.300 | 0.750 | 0.83 |
| 2N6770 | 48 | 750 | 15 | 12.0 | 500 | | 0.400 | 1.000 | 0.83 |

See notes on next page.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center, Columbus, ATTN: DSCC-VAC, Post Office Box 3990, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.3 Maximum ratings - Continued.

(1) Derate linearly, 1.2 W/°C for $T_C$ > +25° C. $P_T = T_J$ max - $T_C$. / $R\theta JC$

(2) $I_D = \frac{\sqrt{T_J \text{ max} - T_C}}{R_{\theta JC} + R_{DS(on)}}$ at $T_J$ max.

(3) $I_{DM}$ = 4 x $I_{D1}$ as calculated in note 2.

1.4 Primary electrical characteristics at $T_C$ = +25°C.

| Type | Min $V_{(BR)DSS}$ $V_{GS}$ = 0V $I_D$ = 1 mA dc | $I_{AR}$ (1) | $E_{AS}$ | $E_{AR}$ | Max $r_{DS(on)}$ $V_{GS}$ = 10 Vdc $I_D$ = ID2 | $V_{GSth1}$ $V_{DS} \geq V_{GS}$ $I_D$ = 0.25 mA | | Max $I_{DSS1}$ VGS = 0 V $V_{DS}$ = 80 percent of rated $V_{DS}$ |
|---|---|---|---|---|---|---|---|---|
| | Vdc | A | mJ | mJ | Ω | min | max | µAdc |
| 2N6764 | 100 | 38.0 | 150 | 15.0 | 0.055 | 2.0 | 4.0 | 25 |
| 2N6766 | 200 | 30.0 | 500 | 15.0 | 0.085 | 2.0 | 4.0 | 25 |
| 2N6768 | 400 | 14.0 | 700 | 15.0 | 0.3 | 2.0 | 4.0 | 25 |
| 2N6770 | 500 | 12.0 | 750 | 15.0 | 0.4 | 2.0 | 4.0 | 25 |

(1) Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

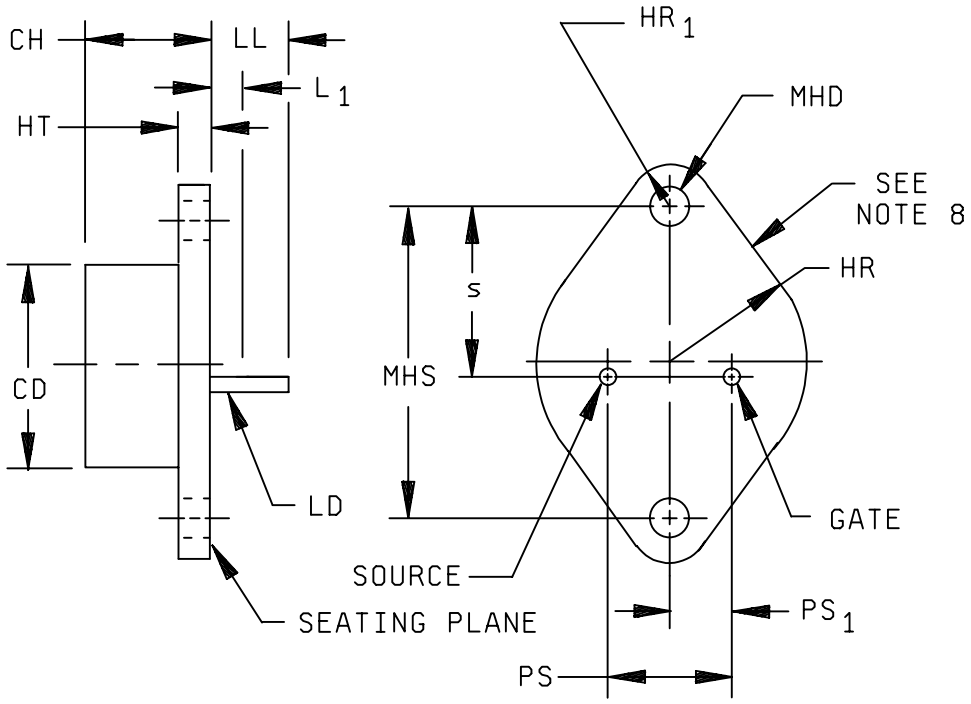
DEPARTMENT OF DEFENSE

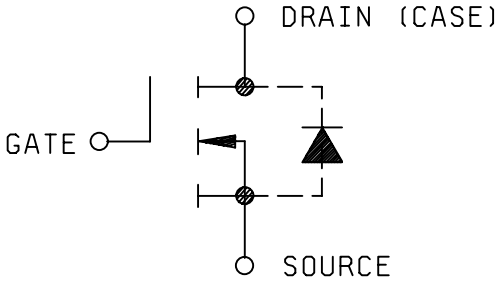
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Document Automation and Production Services (DAPS), Building 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)



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CIRCUIT SCHEMATIC

FIGURE 1. Physical dimensions of transistor types 2N6764 and 2N6766, TO-204AE; for types 2N6768 and 2N6770, TO-204AA.



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| Dimensions | | | | | |
|---|---|---|---|---|---|
| Ltr | Inches | | Millimeter | | Notes |
| | Min | Max | Min | Max | |
| CD | | .875 | | 22.23 | |
| CH | .250 | .360 | 6.35 | 9.15 | |
| HR | .495 | .525 | 12.57 | 13.3 | |
| $HR_1$ | .131 | .188 | 3.33 | 4.78 | |
| HT | .060 | .135 | 1.52 | 3.43 | |
| LD | .057 | .063 | 1.45 | 1.60 | 5 |
| | .038 | .043 | 0.97 | 1.10 | 6 |
| LL | .312 | .500 | 7.92 | 12.70 | |
| $L_1$ | | .050 | | 1.27 | 3 |
| MHD | .151 | .161 | 3.84 | 4.09 | 7 |
| MHS | 1.177 | 1.197 | 29.90 | 30.04 | |
| PS | .420 | .440 | 10.67 | 11.18 | |
| $PS_1$ | .205 | .225 | 5.21 | 5.72 | |
| s | .655 | .675 | 16.64 | 17.15 | |

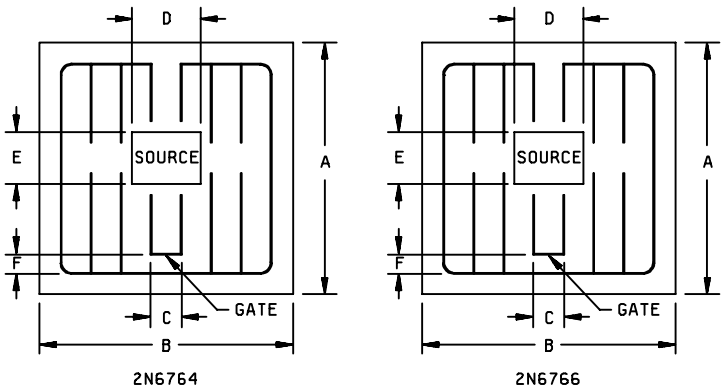
NOTES:

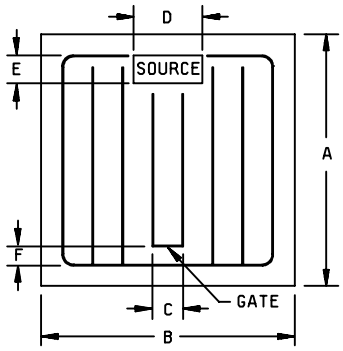
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. These dimensions shall be measured at points .050 inch (1.27 mm) and .055 inch (1.40 mm) below the seating plane. When gauge is not used, measurement will be made at the seating plane.
4. The seating plane of the header shall be flat within .001 inch (0.03 mm) concave to .004 inch (0.10 mm) convex inside a .930 inch (23.62 mm) diameter circle on the center of the header and flat within .001 inch (0.03 mm) concave to .006 inch (0.15 mm) convex overall.
5. These dimensions pertain to the 2N6764 and 2N6766 types.
6. These dimensions pertain to the 2N6768 and 2N6770 types.
7. Mounting holes shall be deburred on the seating plane side.
8. Drain is electrically connected to the case.

FIGURE 1. Physical dimensions of transistor types 2N6764 and 2N6766 TO-204AE; for types 2N6768 and 2N6770, TO-204AA - Continued.



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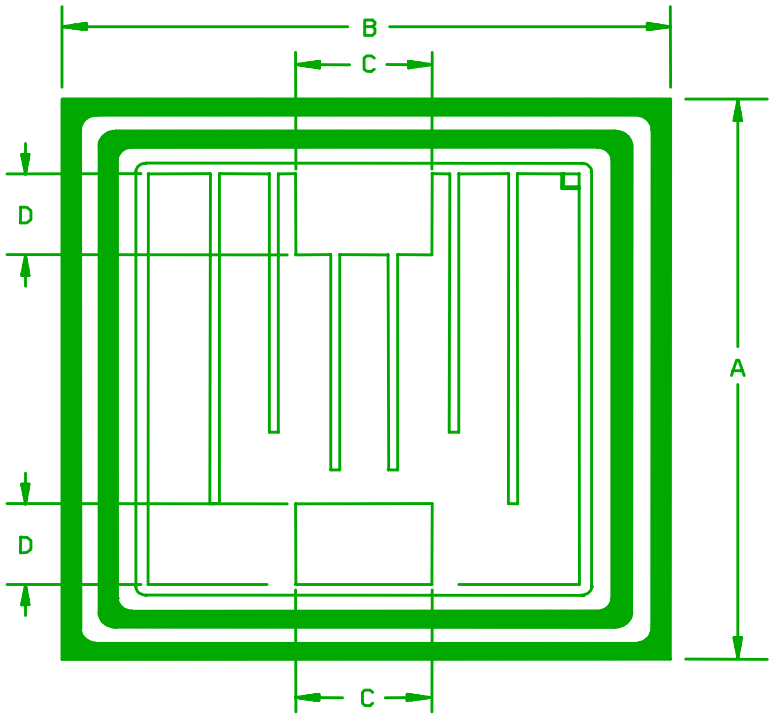

| Ltr | Dimensions 2N6764 and 2N6766 | | | | Dimensions 2N6768 and 2N6770 | | | |
|---|---|---|---|---|---|---|---|---|
| | Inches | | Millimeters | | Inches | | Millimeters | |
| | Min | Max | Min | Max | Min | Max | Min | Max |
| A | .252 | .262 | 6.40 | 6.65 | .252 | .262 | 6.40 | 6.65 |
| B | .252 | .262 | 6.40 | 6.65 | .252 | .262 | 6.40 | 6.65 |
| C | .027 | .037 | 0.69 | 0.94 | .025 | .035 | 0.64 | 0.89 |
| D | .012 | .022 | 0.30 | 0.56 | .043 | .053 | 1.09 | 1.35 |
| E | .057 | .067 | 1.45 | 1.70 | .032 | .042 | 0.81 | 1.07 |
| F | .013 | .023 | 0.33 | 0.58 | .015 | .025 | 0.38 | 0.64 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ± .005 inch (0.13 mm).
4. The physical characteristics of the die thickness are .0187 inch (0.474 mm). The back metals are chromium, nickel and silver. The top metal is aluminum and the back contact is the drain.

FIGURE 2. JANHC and JANKC A-version die dimensions.



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| Dimensions | | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| A | .259 | .269 | 6.58 | 6.83 |
| B | .253 | .263 | 6.43 | 6.68 |
| C | .065 | .075 | 1.65 | 1.91 |
| D | .045 | .055 | 1.14 | 1.40 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ± .005 inch (0.13 mm).
4. The physical characteristics of the die thickness are .014 inch (0.36 mm). The back metals are nickel, aluminum and titanium. The top metal is aluminum and the back contact is the drain.

FIGURE 3. JANHC and JANKC B-version die dimensions.



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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-19500.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figures 1, 2 and 3.

3.4.1 Lead material and finish. Lead material shall be Kovar or Alloy 52; a copper core or a plated core is permitted. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750 and herein.

3.4.2 Construction. These devices shall be constructed in a manner and using materials which enable the devices to meet the applicable requirements of MIL-PRF-19500 and herein.

3.4.3 Internal construction. Multiple chip construction is not permitted to meet the requirements of this specification.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrostatic discharge protection. The devices covered by this specification require electrostatic protection.

3.6.1 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of static discharge. The following handling practices are recommended (see 3.5)

a. Devices shall be handled on benches with conductive and grounded surface.

b. Ground test equipment, tools and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care shall be exercised during test and troubleshooting to apply not more than maximum rated voltage to any lead.

h. Gate shall be terminated to source, $R \leq 100$ k, whenever bias voltage is to be applied drain to source.

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4 and table I.

3.8 Electrical test requirements. The electrical test requirements shall be subgroups specified in 4.4.2 and 4.4.3.



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3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 JANHC and JANKC devices. Qualification for JANHC and JANKC devices shall be as specified in MIL-PRF-19500, appendix G.

4.3 Screening (JANS, JANTXV and JANTX levels only). Screening shall be in accordance with appendix E, table IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I shall not be acceptable.

| Screen (see MIL-PRF-19500, appendix E, table IV | Measurements | |
|---|---|---|
| | JANS level | JANTX and JANTXV |
| (1) | Gate stress test (see 4.5.5) | Gate stress test (see 4.5.5). |
| (1) (2) | Method 3470 of MIL-STD-750, (see 4.5.4) | Method 3470 of MIL-STD-750, (see 4.5.4) |
| (1) (3) | Method 3161 of MIL-STD-750, (see 4.5.3) | Method 3161 of MIL-STD-750, (see 4.5.3) |
| 9 | $I_{GSS1}$, $I_{DSS1}$ | |
| 10 | Method 1042 of MIL-STD-750, test condition B | Method 1042 of MIL-STD-750, test condition B |
| 11 | $I_{GSS1}$, $I_{DSS1}$, $r_{DS(on)1}$.<br>$V_{GS(th)1}$ of subgroup 2 of table I herein.<br>$\Delta I_{GSS1}$ = ± 20 nA dc or ± 100 percent of initial value, whichever is greater.<br>$\Delta I_{DSS1}$ = ± 25 µA dc or ± 100 percent of initial value, whichever is greater | $I_{GSS1}$, $I_{DSS1}$, $r_{DS(on)1}$.<br>$V_{GS(th)1}$ of subgroup 2 of table I herein |
| 12 | Method 1042 of MIL-STD-750, test condition A or t = 240 hours | Method 1042 of MIL-STD-750, test condition A, t = 48 hours minimum at $T_A$ = +175° C minimum. |
| 13 | Subgroups 2 and 3 of table I.<br>$\Delta I_{GSS1}$ = ± 20 nA dc or ± 100 percent of initial value whichever is greater.<br>$\Delta I_{DSS1}$ = ± 25 µA dc or ± 100 percent of initial value whichever is greater.<br>$\Delta r_{DS(on)1}$ = ± 20 percent of initial value<br>$\Delta V_{GS(th)1}$ = ± 20 percent of initial value. | Subgroup 2 of table I herein.<br>$\Delta I_{GSS1}$ = ± 20 nA dc or ± 100 percent of initial value whichever is greater.<br>$\Delta I_{DSS1}$ = ± 25 µA dc or ± 100 percent of initial value whichever is greater.<br>$\Delta r_{DS(on)1}$ = ± 20 percent of initial value.<br>$\Delta V_{GS(th)1}$ = ± 20 percent of initial value. |

(1) Shall be performed anytime before screen 10.
(2) This test method in no way implies a repetitive avalanche energy rating.
(3) This test need not be performed in group A when performed as a screen.



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4.3.1 Screening (JANHC and JANKC). Screening shall be in accordance with appendix E, table IV of MIL-PRF 19500. As a minimum, die shall be 100 percent probed in accordance with group A, subgroup 2 except test current shall not exceed 20 A.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein. Alternate flow is allowed for conformance inspection in accordance with figure 4 of MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with appendix E, table V of MIL-PRF-19500. End-point electrical and delta measurements shall be in accordance with the applicable tests of table III herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VIa (JANS) and table VIb (JANTX, JANTXV and JAN) of MIL-PRF-19500 and as follows. End-point electrical and delta measurements shall be in accordance with the applicable steps of table III herein.

4.4.2.1 Group B inspection, appendix E, table VIa (JANS) of MIL-PRF-19500.

| Subgroup | Method | Conditions |
|---|---|---|
| B3 | 1051 | Test condition G. |
| B4 | 1042 | Test condition D; the heating cycle shall be 1 minimum for 2,000 cycles. |
| B5 | 1042 | Test condition A; $V_{DS}$ = rated $V_{DS}$ (see 1.3), $T_A$ = 175° C, t = 120 hours minimum, read and record $V_{BR(DSS)}$ (pre and post) at $I_D$ = 1 mA, read and record $I_{DSS}$ (pre and post), (see table III). |
| B5 | 1042 | Test condition B; $V_{GS}$ = rated $V_{GS}$ (see 1.3), $T_A$ = 175° C, t = 24 hours minimum. |
| B6 | 3161 | See 4.5.2. |

4.4.2.2 Group B inspection, appendix E, table VIb (JAN, JANTX and JANTXV) of MIL-PRF-19500.

| Subgroup | Method | Conditions |
|---|---|---|
| B2 | 1051 | Test condition G. |
| B3 | 1042 | Test condition D, 2,000 cycles. The heating cycle shall be 1 minute minimum. |
| B5, B6 | | Not applicable. |

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VII of MIL-PRF-19500. End-point electrical and delta measurements shall be in accordance with the applicable steps of table III herein.



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4.4.3.1 Group C inspection, appendix E, table VII of MIL-PRF-19500.

| Subgroup | Method | Conditions |
|---|---|---|
| C2 | 2036 | Test condition A; weight = 10 lbs, t = 15 seconds. |
| C6 | 1042 | Test condition D; 6,000 cycles minimum. The heating cycle shall be 1 minute minimum. |

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750. $R_{\theta JC}$max = 0.83° C /W.

a. $I_M$ measuring current ..... 10 mA.

b. $I_H$ drain heating current ..... 4 A minimum.

c. $t_H$ heating time ..... Steady-state (see method 3161 of MIL-STD-750 for definition).

d. $V_H$ drain-source heating voltage ..... 25 V minimum.

e. $t_{MD}$ measurement time delay ..... 30 to 60 µs.

f. $t_{SW}$ sample window time ..... 10 µs (max).

4.5.3 Thermal response ($\Delta V_{SD}$ measurements). The delta $V_{SD}$ measurements shall be performed with method 3161 of MIL-STD-750. The delta $V_{SD}$ conditions ($I_H$ and $V_H$) and maximum limit shall be derived by each vendor from the thermal response curves (see figure 4). The read and record delta $V_{SD}$ measurements and conditions for each device in the qualification lot shall be submitted in the qualification report. The chosen delta $V_{SD}$ shall be considered final after the manufacturer has had the opportunity to test five consecutive lots. The following parameter measurements shall apply:

a. $I_M$ measuring current ..... 10 mA.

b. $I_H$ drain heating current ..... 4 A minimum.

c. $t_H$ heating time ..... 100 ms.

d. $V_H$ drain-source heating voltage ..... 25 V minimum.

e. $t_{MD}$ measurement time delay ..... 30 to 60 µs.

f. $t_{SW}$ sample window time ..... 10 µs (max).



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4.5.4 Single pulsed unclamped inductive switching.

a. Peak current ..... $I_{D1}$.

b. Peak gate voltage, $V_{GS}$ ..... 10 V.

c. Gate to source resistor, $R_{GS}$..... $25 \leq R_g \leq 200$ ohms.

d. Initial case temperature ..... +25°C, +10°C, -5°C.

e. Inductance, L ..... $\left[\frac{2E_{AS}}{(I_{D1})^2}\right]\left[\frac{(V_{BR}-V_{DD})}{V_{BR}}\right]$ mH minimum.

f. Number of pulses to be applied..... 1 pulse minimum.

g. Supply voltage ($V_{DD}$) ..... 50 V, (25 V for devices with minimum $V_{(BR)DSS}$ of 100 V).



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TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical inspection | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Breakdown voltage drain to source | 3407 | $V_{GS}$ = 0 V dc; $I_D$ = 1 mA dc, bias condition C | $V_{(BR)DSS}$ | | | V dc |
| 2N6764 | | | | 100 | | |
| 2N6766 | | | | 200 | | |
| 2N6768 | | | | 400 | | |
| 2N6770 | | | | 500 | | |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$; $I_D$ = 0.25 mA dc | $V_{GS(th)1}$ | 2.0 | 4.0 | V dc |
| Gate current | 3411 | $V_{GS}$ = +20 and -20 V dc; bias condition C, $V_{DS}$ = 0 | $I_{GSS1}$ | | 100 | nA dc |
| Drain current | 3413 | $V_{GS}$ = 0 V dc; $V_{DS}$ = 80 percent of rated $V_{DS}$, bias condition C | $I_{DSS1}$ | | 25 | µA dc |
| Static drain to source on-state resistance | 3421 | $V_{GS}$ = 10 V dc, pulsed (see 4.5.1), condition A $I_D$ = rated $I_{D2}$ (see 1.3) $T_C$ = +25° C. | $r_{DS(on)1}$ | | | Ω |
| 2N6764 | | | | | 0.055 | |
| 2N6766 | | | | | 0.085 | |
| 2N6768 | | | | | 0.3 | |
| 2N6770 | | | | | 0.4 | |
| Static drain to source on-state resistance | 3421 | $V_{GS}$ = 10 V dc, pulsed (see 4.5.1), condition A $I_D$ = rated $I_{D1}$ (see 1.3) | $r_{DS(on)2}$ | | | Ω |
| 2N6764 | | | | | 0.065 | |
| 2N6766 | | | | | 0.09 | |
| 2N6768 | | | | | 0.40 | |
| 2N6770 | | | | | 0.50 | |
| Forward voltage (source-drain diode) | 4011 | Pulsed (see 4.5.1) $V_{GS}$ = 0 V, $I_D$ = $I_{D1}$ | $V_{SD}$ | | | V dc |
| 2N6764 | | | | | 1.9 | |
| 2N6766 | | | | | 1.9 | |
| 2N6768 | | | | | 1.7 | |
| 2N6770 | | | | | 1.7 | |

See footnote at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 3 | | | | | | |
| High temperature operation: | | $T_C$ = +125° C | | | | |
| Gate current | 3411 | Bias condition C; $V_{GS}$ = +20 and -20 V dc $V_{DS}$ = 0 V dc | $I_{GSS2}$ | | 200 | nA dc |
| Drain current | 3413 | Bias condition C; $V_{GS}$ = 0 V dc | | | | |
| | | $V_{DS}$ = 100 percent of rated $V_{DS}$ | $I_{DSS2}$ | | 1.0 | mA dc |
| | | $V_{DS}$ = 80 percent of rated $V_{DS}$ | $I_{DSS3}$ | | 0.25 | mA dc |
| Static drain to source on-state resistance | 3421 | $V_{GS}$ = 10 V dc pulsed (see 4.5.1) $I_D$ = rated $I_{D2}$ (see 1.3) | $r_{DS(on)3}$ | | | Ω |
| 2N6764 | | | | | 0.094 | |
| 2N6766 | | | | | 0.153 | |
| 2N6768 | | | | | 0.66 | |
| 2N6770 | | | | | 0.88 | |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$; $I_D$ = 0.25 mA dc | $V_{GS(th)2}$ | 1.0 | | V dc |
| Low temperature operation: | | $T_C$ = -55° C | | | | |
| Gate to source voltage (threshold) | | $V_{DS} \geq V_{GS}$; $I_D$ = 0.25 mA dc | $V_{GS(th)3}$ | | 5.0 | V dc |
| Subgroup 4 | | | | | | |
| Switching time test | 3472 | $I_D$ = rated $I_{D1}$ (see 1.3) $V_{GS}$ = 10 V dc Gate drive impedance = 2.35 Ω $V_{DD}$ = 0.5 $V_{BR(DSS)}$ | | | | |
| Turn-on delay time | | | $t_{d(on)}$ | | 35 | ns |
| Rise time | | | $t_r$ | | 190 | ns |
| Turn-off delay time | | | $t_{d(off)}$ | | 170 | ns |
| Fall time | | | $t_f$ | | 130 | ns |

See footnote at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 5 | | | | | | |
| Safe operating area test | 3474 | See figure 5, $V_{DS}$ = 80 percent of rated $V_{BR(DSS)}$, $t_p$ = 10 ms, $V_{DS}$ = 200 V max. | | | | |
| Electrical measurements | | See table III, steps, 1, 2, 3, 4, 5 6 and 7. | | | | |
| Subgroup 6 | | | | | | |
| Not applicable | | | | | | |
| Subgroup 7 | | | | | | |
| Gate charge | 3471 | | | | | |
| On-state gate charge | | Bias condition B | $Q_g(on)$ | | | nC |
| 2N6764 | | | | | 125 | |
| 2N6766 | | | | | 115 | |
| 2N6768 | | | | | 110 | |
| 2N6770 | | | | | 120 | |
| Gate to source charge | | | $Q_{gs}$ | | | nC |
| 2N6764 | | | | | 22 | |
| 2N6766 | | | | | 22 | |
| 2N6768 | | | | | 18 | |
| 2N6770 | | | | | 19 | |
| Gate to drain charge | | | $Q_{gd}$ | | | nC |
| 2N6764 | | | | | 65 | |
| 2N6766 | | | | | 60 | |
| 2N6768 | | | | | 65 | |
| 2N6770 | | | | | 70 | |
| Reverse recovery time | 3473 | | $t_{rr}$ | | | nC |
| 2N6764 | | di/dt = 100 A/µs, $V_{DD} \leq$ 30 V dc, $I_D = I_{D1}$ | | | 500 | |
| 2N6766 | | | | | 950 | |
| 2N6768 | | | | | 1,200 | |
| 2N6770 | | | | | 1,600 | |

1/ For sampling plan, see MIL-PRF-19500.



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TABLE II. Group E inspection (all quality levels) for qualification only.

| Inspection 1/ | MIL-STD-750 | | Qualification and large lot quality conformance inspection |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 45 devices, c = 0 |
| Temperature cycling | 1051 | Test condition G, 500 cycles | |
| Hermetic seal | 1071 | | |
| Fine leak<br>Gross leak | | | |
| Electrical measurements | | See table III, steps, 1, 2, 3, 4, 5, 6 and 7. | |
| Subgroup 2 1/ | | | 45 devices, c = 0 |
| Steady-state reverse bias | 1042 | Condition A; 1,000 hours | |
| Electrical measurements | | See table III, steps, 1, 2, 3, 4, 5, 6 and 7. | |
| Steady-state gate bias | 1042 | Condition B, 1,000 hours | |
| Electrical measurements | | See table III, steps, 1, 2, 3, 4, 5, 6 and 7. | |
| Subgroup 3 | | | |
| Not applicable | | | |
| Subgroup 4 | | | 5 devices, c = 0 |
| Thermal resistance | 3161 | $R_{\theta JC}$ = 0.83° C/W max. (see 4.5.2) | |
| Subgroup 5 | | | 5 devices, c = 0 |
| Barometric pressure (reduced)<br>400 V and 500 V only | 1001 | Test condition C;<br>$I_{(ISO)}$ = .25 mA (max),<br>$V_{(ISO)} = V_{DS}$ | |
| Subgroup 6 | | | |
| Not applicable | | | |
| Subgroup 7 | | | 5 devices, c = 0 |
| Repetive avalanche energy | 3469 | $I_{AR} = I_D$; $V_{GS}$ = 10 V;<br>$2.5 \leq R_{GS} \leq 200$ ohms;<br>$T_J$ = 150°C +10, -0 °C;<br>Inductance = $\left[\frac{2E_{AR}}{(I_{D1})^2}\right]\left[\frac{V_{BR}-V_{DD}}{V_{BR}}\right]$ $mH$ min<br>Number of pulses to be applied = $3.6 \times 10^8$;<br>($V_{DD}$) = 50 V;<br>time in avalanche = 2 µs minimum, 20 µs maximum;<br>f = 1 KHz | |

1/ A separate sample may be pulled for each test.



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TABLE III. Groups A, B, C and E electrical measurements. 1/ 2/ 3/

| Step | Inspection | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | | Method | Conditions | | Min | Max | |
| 1. | Breakdown voltage drain to source<br><br>2N6764<br>2N6766<br>2N6768<br>2N6770 | 3407 | $V_{GS}$ = 0, $I_D$ = 1 mA dc bias condition C; | $V_{(BR)DSS}$ | <br><br>100<br>200<br>400<br>500 | | V dc |
| 2. | Gate to source voltage (threshold) | 3404 | $V_{DS} \geq V_{GS}$<br>$I_D$ =0.25 mA dc | $V_{GS(th)1}$ | 2.0 | 4.0 | V dc |
| 3. | Gate current | 3411 | $V_{GS}$ = 20<br>Bias condition C; | $I_{GSS1}$ | | 100 | nA dc |
| 4. | Saturation voltage and resistance | 3413 | $V_{GS}$ = 0 $V_{DS}$ = 80 percent of rated $V_D$, bias condition C; | $I_{DSS1}$ | | 25 | µA dc |
| 5. | Static drain to source on-state resistance<br><br>2N6764<br>2N6766<br>2N6768<br>2N6770 | 3421 | $V_{GS}$ = 10 V dc condition A, pulsed (see 4.5.1). $I_D$ = $I_{D2}$ | $r_{DS(on)1}$ | | <br><br>0.055<br>0.085<br>0.3<br>0.4 | ohms |
| 6. | Static drain to source on-state resistance<br><br>2N6764<br>2N6766<br>2N6768<br>2N6770 | 3421 | $V_{GS}$ = 10 V dc condition A, pulsed (see 4.5.1). $I_D$ = $I_{D1}$ | $r_{DS(on)2}$ | | <br><br>0.065<br>0.090<br>0.400<br>0.500 | ohms |
| 7. | Forward voltage (source-drain diode)<br><br>2N6764<br>2N6766<br>2N6768<br>2N6770 | 4011 | $V_{GS}$ = 0 V dc;<br>$I_D$ = $I_{D1}$<br>pulsed (see 4.5.1) | $V_{SD}$ | | <br><br>1.9<br>1.9<br>1.7<br>1.7 | V |
| 8. | Thermal response | 3131 | See 4.5.3 | $\Delta V_{SD}$ | | | |

See footnotes on next page



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TABLE III. Groups A, B, C and E electrical measurements. 1/ 2/ 3/ - Continued.

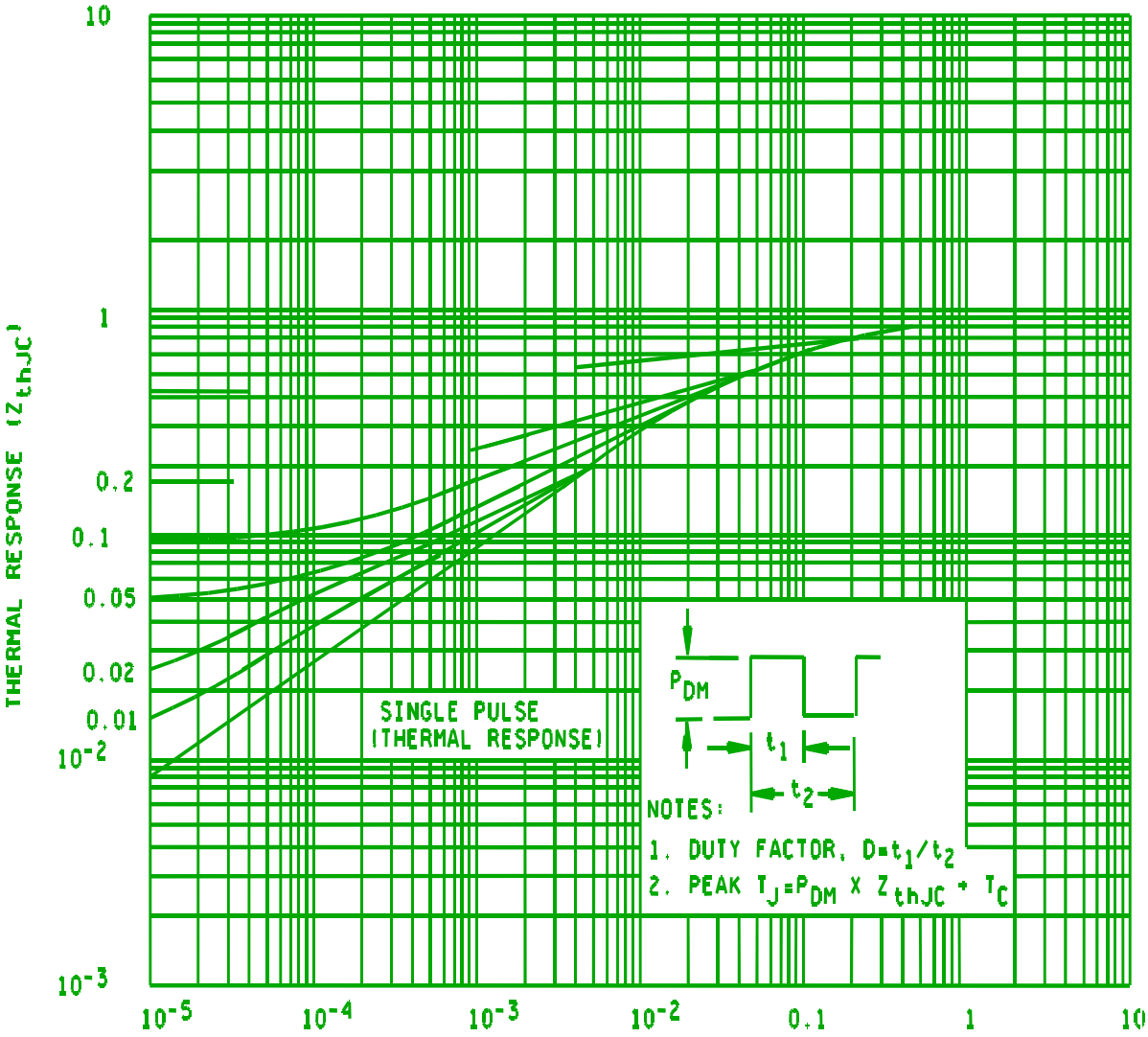
1/ The electrical measurements for appendix E, table VIa (JANS) of MIL-PRF-19500 are as follows:
   a. Subgroup 3, table III, steps 1, 2, 3, 4, 5, 6 and 7.
   b. Subgroup 4, table III, steps 1, 2, 3, 4, 5, 6, 7 and 8.
   c. Subgroup 5, table III, condition A, steps 1, 2, 3, 4, 5, 6 and 7. No more than 15 percent of the sample shall be permitted to have a $\Delta V_{BR(DSS)}$ shift of more than 10 percent and $\Delta I_{DSS}$ greater than 50 µA. Subgroup 5, table III, condition B, steps 1, 2, 3, 4, 5, 6 and 7.

2/ The electrical measurements for appendix E, table VIb (JANTX and JANTXV) of MIL-PRF-19500 are as follows:
   a. Subgroup 2, table III, steps 1, 2, 3, 4, 5, 6 and 7.
   b. Subgroup 3, table III, steps 1, 2, 3, 4, 5, 6, 7 and 8.

3/ The electrical measurements for appendix E, table VII of MIL-PRF-19500 are as follows:
   a. Subgroup 2, table III, steps 1, 2, 3, 4, 5, 6 and 7.
   b. Subgroup 3, table III, steps 1, 2, 3, 4, 5, 6 and 7.
   c. Subgroup 6, table III, steps 1, 2, 3, 4, 5, 6, 7 and 8.



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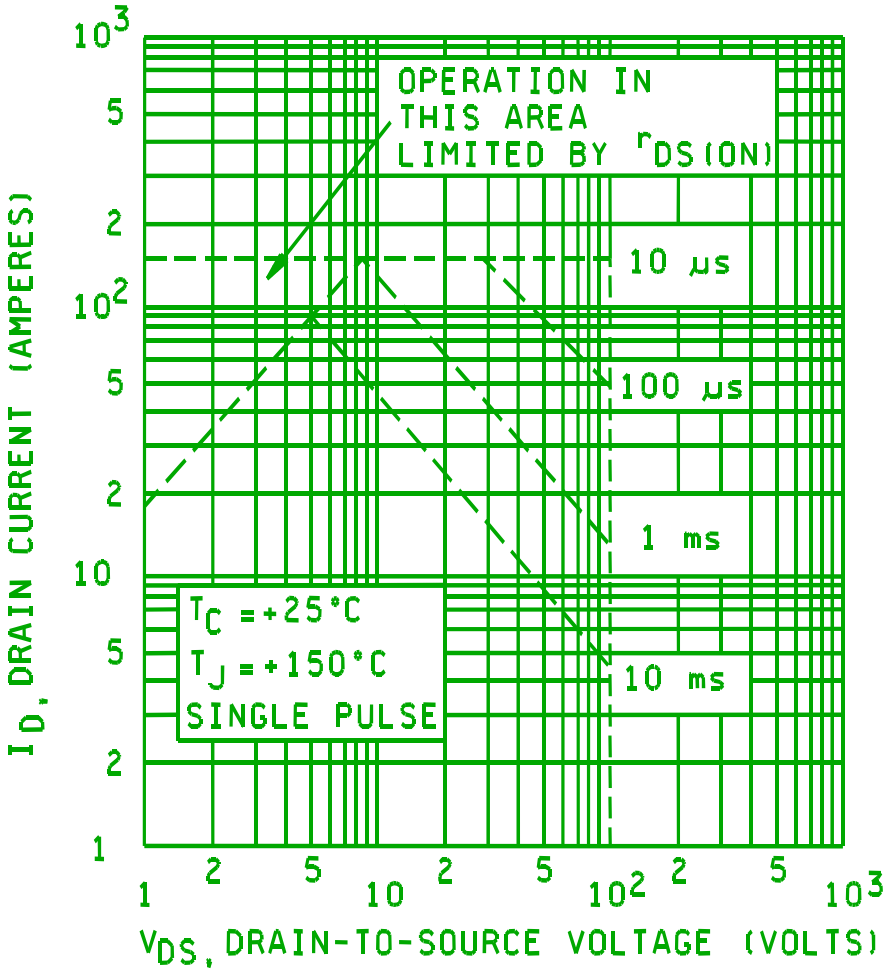



$t_1$ RECTANGLE PULSE DURATION (SECONDS)

FIGURE 4. Thermal response curves.



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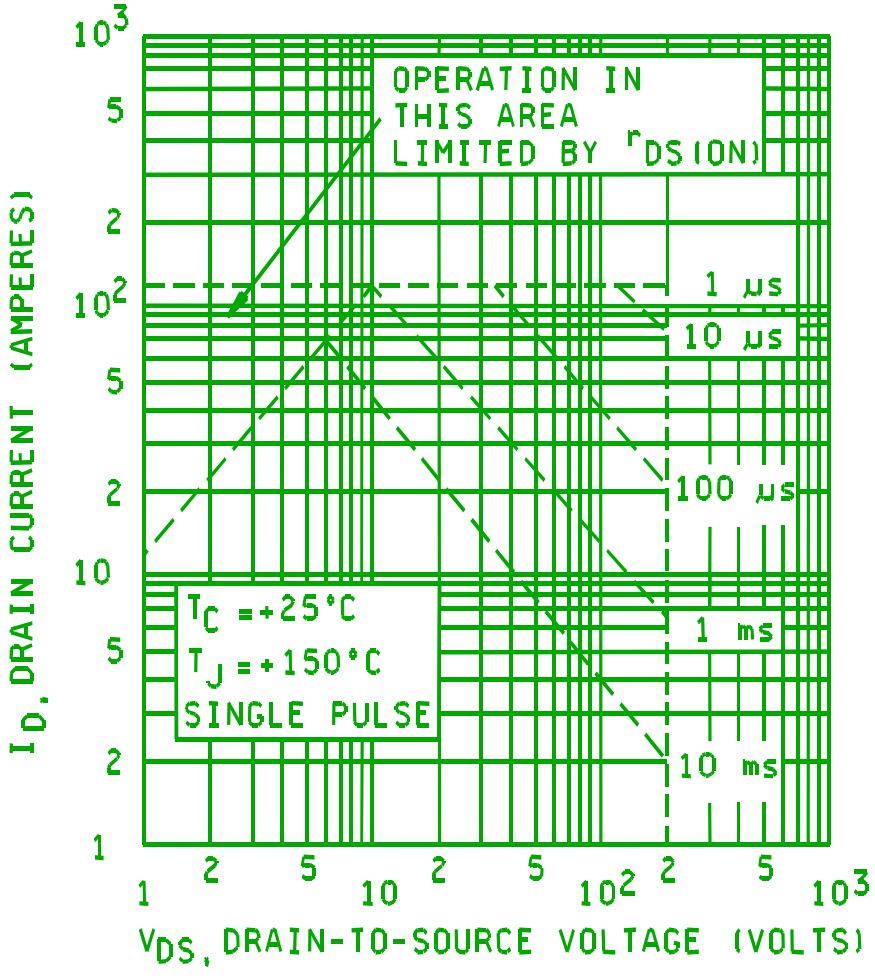



2N6764

FIGURE 5. Safe operating area graph.



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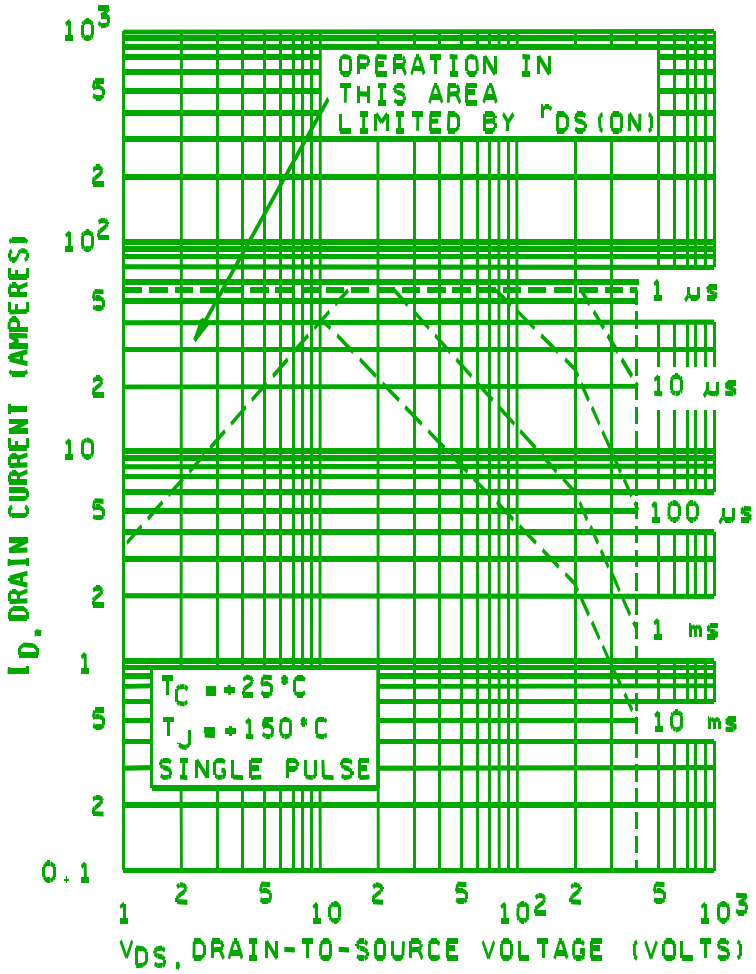



2N6766

FIGURE 5. Safe operating area graph - Continued.



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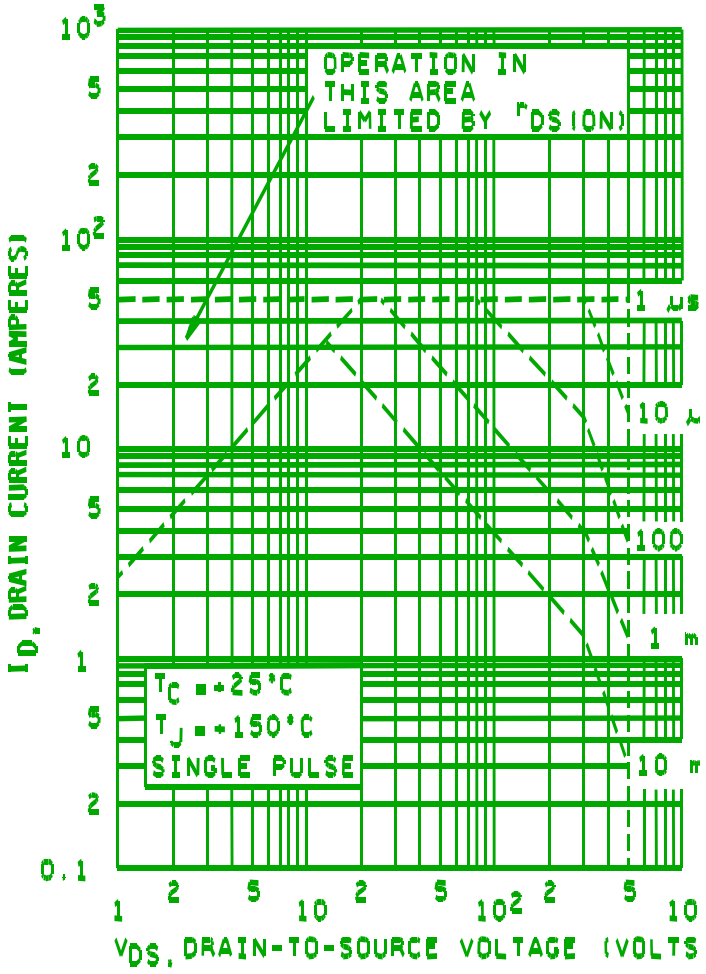



2N6768

FIGURE 5. Safe operating area graph - Continued.



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2N6770

FIGURE 5. Safe operating area graph - Continued.



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5. PACKAGING

5.1 Packaging. Packaging shall prevent mechanical damage of the devices during shipping and handling and shall not be detrimental to the device. When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. The acquisition requirements are as specified in MIL- PRF-19500.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from: Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43216-5000.

6.4 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's PIN. This information in no way implies that manufacturer's PINs are suitable as a substitute for the military Part or Identifying Number (PIN).

| PIN | Manufacturer’s CAGE code | Manufacturer’s and user’s PIN |
|---|---|---|
| 2N6764 | 59993, 18722 | IRF150, IRF151, IRF152, IRF153 |
| 2N6766 | 59993, 18722 | IRF250, IRF251, IRF252, IRF253 |
| 2N6768 | 59993, 18722 | IRF350, IRF351, IRF352, IRF353 |
| 2N6770 | 59993, 18722 | IRF450, IRF451, IRF452, IRF453 |

6.5 Replacement data. JANTX devices shall be a direct one way replacement for JAN devices (example: JANTX2N6764 for JAN2N6764).



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6.6 Suppliers of JANC die. The qualified JANC suppliers with the applicable letter version (example JANHCAM2N6764) will be identified on the QPL.

| JANC ordering inforation | | |
|---|---|---|
| PIN | Manufacturer | |
| | 59993 | 18722 |
| 2N6764 | JANHCA2N6764<br>JANTXHCA2N6764<br>JANTXVHCA2N6764<br>JANSHCA2N6764 | JANHCB2N6764<br>JANTXHCB2N764<br>JANTXVHCB2N6764<br>JANSHCB2N6764 |
| 2N6766 | JANHCA2N6766<br>JANTXHCA2N6766<br>JANTXVHCA2N6766<br>JANSHCA2N6766 | JANHCB2N6766<br>JANTXHCB2N766<br>JANTXVHCB2N6766<br>JANSHCB2N6766 |
| 2N6768 | JANHCA2N6768<br>JANTXHCA2N6768<br>JANTXVHCA2N6768<br>JANSHCA2N6768 | JANHCB2N6768<br>JANTXHCB2N768<br>JANTXVHCB2N6768<br>JANSHCB2N6768 |
| 2N6770 | JANHCA2N6770<br>JANTXHCA2N6770<br>JANTXVHCA2N6770<br>JANSHCA2N6770 | JANHCB2N6770<br>JANTXHCB2N770<br>JANTXVHCB2N6770<br>JANSHCB2N6770 |

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy -NW
Air Force - 11
DLA - CC

Review activities:
NAVY - TD
Air Force - 19, 70, 99

Preparing activity:
DLA - CC

(Project 5961-2365)



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## STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| **I RECOMMEND A CHANGE:** | 1. DOCUMENT NUMBER<br>MIL-PRF-19500/543F | 2. DOCUMENT DATE<br>7 September 2001 |
|---|---|---|

3. **DOCUMENT TITLE**
SEMICONDUCTOR DEVICE, FIELD EFFECT TRANSISTORS, N-CHANNEL, SILICON REPETITIVE AVALANCHE TYPES 2N6764, 2N6766, 2N6768, 2N6770, JAN, JANTX, JANTXV, JANS, JANHC and JANKC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>COMMERCIAL<br>DSN<br>FAX<br>EMAIL | 7. DATE SUBMITTED |

8. PREPARING ACTIVITY

| a. Point of Contact<br>Alan Barone | b. TELEPHONE<br>Commercial<br>614-692-0510 | DSN<br>850-0510 | FAX<br>614-692-6939 | EMAIL<br>alan.barone@dscc.dla.mil |
|---|---|---|---|---|
| c. ADDRESS<br>Defense Supply Center Columbus<br>ATTN: DSCC-VAC<br>P.O. Box 3990<br>Columbus, OH 43216-5000 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Standardization Program Office (DLSC-LM)<br>8725 John J. Kingman, Suite 2533<br>Fort Belvoir, VA 22060-6221<br>Telephone (703) 767-6888 DSN 427-6888 | | | |

**DD Form 1426, Feb 1999 (EG)** Previous editions are obsolete WHS/DIOR, Feb 99

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